

LIST KERJAAN MINGGUAN

Timeline Proyek Sistem GLD

20 April 2026 – 7 Juni 2026 | 7 Minggu

Cluster Head

Gas Leak Detector System

Skenario 1

Jaringan Normal

Proses inisialisasi & pembentukan topologi mesh



STEP 1

CH3 / GATEWAY Aktif

Address:	CH3 = 0x0003
U1 RX:	921 MHz (discovery + CH1 direct)
U3 RX:	922 MHz (CH2 data)
WiFi:	Connecting... → GAGAL → Retry
MQTT:	OFFLINE → OK ✓
GW Siap:	cluster=0x0003
HB:	rx=0 ok=0 WiFi=OK MQTT=OK

```
16:02:57.144 ->
16:02:57.144 -> === CH3 / GATEWAY (0x0003) ===
16:02:57.144 -> U1@921MHz: discovery+CH1direct | U3@922MHz: data CH2
16:02:57.144 -> !!! NYALAKAN INI PERTAMA !!!
16:02:57.144 ->
16:02:57.144 -> [WiFi] aqms-modem-001.....
16:03:12.199 -> [WiFi] GAGAL
16:03:12.199 -> [U1-921] Init... OK
16:03:12.284 -> [U3-922] Init... OK
16:03:12.318 -> [U1] Listening 921MHz (discovery + CH1 direct)
16:03:12.318 -> [U3] Listening 922MHz (CH2 data)
16:03:13.221 ->
16:03:13.221 -> [GW] Siap | cluster=0x0003 | MQTT:OFFLINE
16:03:13.221 ->
16:03:15.783 -> [MQTT] OK
16:03:24.933 -> [HB|30s] rx=0 ok=0 WiFi=OK MQTT=OK
```

STEP 2

CH2 Aktif & Melakukan Config

Address: CH2 = 0x0002

U1 RX: 921 MHz (dari CH1)

U3 TX: 922 MHz (ke CH3)

Discovery: Broadcast 1/5 @ 922MHz

Response: 0x0003 RSSI=-112.0dBm (2x)

Parent: 0x0003 @ -112.0dBm

CH_HELLO: → 0x0003 → Terkirim ✓

Status: CH2 Siap | parent=0x0003

```
16:04:48.733 -> === CH2 (0x0002) ===
16:04:48.733 -> U1:921MHz RX CH1 | U3:922MHz TX CH3 | ACK+Failover
16:04:48.733 ->
16:04:48.733 -> [PREFS] Loaded parent=0x0003 alt=0x0000
16:04:48.733 -> [U1-921] Init... OK
16:04:48.816 -> [U3-922] Init... OK
16:04:48.862 ->
16:04:48.862 -> [CH_CONFIG] Discovery @ 922MHz...
16:04:48.862 -> [CH_CONFIG] Broadcast 1/5...
16:04:49.320 -> [CH_CONFIG] Listening 5 detik...
16:04:49.919 -> [CH_CONFIG] Resp 0x0003 RSSI=-112.0dBm
16:04:50.646 -> [CH_CONFIG] Resp 0x0003 RSSI=-112.0dBm
16:04:54.328 -> [CH_CONFIG] parent=0x0003(-112.0dBm) alt=0x0000
16:04:54.328 -> [PREFS] Saved parent=0x0003 alt=0x0000
16:04:54.631 -> [CH_HELLO] →0x0003
16:04:55.104 -> [CH_HELLO] Terkirim
16:04:55.104 -> [U1] Listening 921MHz CH1...
16:04:55.104 -> [U3] Listening 922MHz CH3...
16:04:55.104 ->
16:04:55.104 -> [CH2] Siap | parent=0x0003 alt=0x0000
16:04:55.104 ->
```

STEP 3

CH3/GW Merespon Config CH2

Event: [U3 CH_CONFIG] Req dari 0x0002

RSSI: -112.0dBm

Aksi: Respond ke 0x0002 (delay 400ms)

Resp ke-1: @ 16:04:49.871

Resp ke-2: @ 16:04:50.645

HB: rx=1 ok=0 WiFi=OK MQTT=OK

CH_HELLO: [U3] CH=0x0002 RSSI:-112.0dBm

16:04:49.319 -> [U3 CH_CONFIG] Req dari 0x0002 RSSI:-112.0dBm

16:04:49.319 -> [CH_CONFIG] Respond ke 0x0002 (delay 400ms)

16:04:49.871 -> [CH_CONFIG] Resp ke-1

16:04:50.645 -> [CH_CONFIG] Resp ke-2

16:04:54.929 -> [HB|120s] rx=1 ok=0 WiFi=OK MQTT=OK

16:04:55.103 -> [U3 CH_HELLO] CH=0x0002 RSSI:-112.0dBm

Alur Komunikasi:

CH2 → [CH_CONFIG Broadcast] → CH3/GW

CH3/GW → [Resp ke-1, ke-2] → CH2

STEP 4

CH1 Aktif & Melakukan Config

Address:	CH1 = 0x0001
Mode:	STAR:920MHz RX MESH:921MHz TX
Config:	Broadcast 1/5 @ 921MHz
Resp ke-1:	0x0002 RSSI=-95.0dBm
Resp ke-2:	0x0002 RSSI=-94.0dBm (TERBAIK ✓)
Parent:	0x0002 @ -94.0dBm
CH_HELLO:	→ 0x0002 → Terkirim ✓
Status:	CH1 Siap parent=0x0002

```
16:08:00.682 -> === CH1 (0x0001) ===
16:08:00.682 -> STAR:920MHz RX | MESH:921MHz TX | ACK+Failover+PREFS
16:08:00.719 ->
16:08:00.719 -> [PREFS] Loaded parent=0x0002 alt=0x0000
16:08:00.719 -> [U1-920] Init... OK
16:08:00.802 -> [U3-921] Init... OK
16:08:00.836 ->
16:08:00.836 -> [CH_CONFIG] Mencari parent terbaik @ 921MHz...
16:08:00.836 -> [CH_CONFIG] Broadcast 1/5...
16:08:01.305 -> [CH_CONFIG] Listening response 5 detik...
16:08:01.876 -> [CH_CONFIG] Response 0x0002 RSSI=-95.0dBm
16:08:02.647 -> [CH_CONFIG] Response 0x0002 RSSI=-94.0dBm
16:08:06.308 -> [CH_CONFIG] SELESAI: parent=0x0002 (-94.0dBm) alt=0x0000
16:08:06.308 -> [PREFS] Saved parent=0x0002 alt=0x0000
16:08:06.606 -> [CH_HELLO] → 0x0002
16:08:07.091 -> [CH_HELLO] Terkirim
16:08:07.091 -> [U1] Listening 920MHz...
16:08:07.091 -> [U3] Listening 921MHz...
16:08:07.091 ->
16:08:07.091 -> [CH1] Siap | parent=0x0002 alt=0x0000
16:08:07.091 ->
```

STEP 5

CH2 Merespon Config dari CH1

Event: CH_CONFIG Req dari CH=0x0001

RSSI: -100.0dBm

Aksi: Respond ke 0x0001 (delay 400ms)

Resp ke-1: @ 16:08:01.876

Resp ke-2: @ 16:08:02.647

CH_HELLO: dari CH=0x0001 RSSI:-100.0dBm

16:08:01.306 ->

16:08:01.306 -> [CH_CONFIG] Req dari CH=0x0001 RSSI:-100.0dBm

16:08:01.306 -> [CH_CONFIG] Respond ke 0x0001 (delay 400ms)

16:08:01.876 -> [CH_CONFIG] Resp ke-1

16:08:02.647 -> [CH_CONFIG] Resp ke-2

16:08:07.091 -> [CH_HELLO] dari CH=0x0001 RSSI:-100.0dBm

CH2 mencatat CH1 sebagai child node.

CH1 parent=0x0002 ✓ | HELLO Terkirim

CH1 → [Config Req @ 921MHz] → CH2 → [Resp ke-1, ke-2] → CH1 → [CH_HELLO] → CH2

STEP 6

CH3/GW Merespon Config & Hello CH1

Event: [U1 CH_CONFIG] Req dari 0x0001

RSSI: -109.0dBm

Aksi: Respond ke 0x0001 (delay 400ms)

Resp ke-1: @ 16:08:01.876

Resp ke-2: @ 16:08:02.647

CH_HELLO: CH=0x0001 RSSI:-109.0dBm

Dicatat: cluster=0x0001 parent=0x0002

HB: rx=5 ok=0 WiFi=OK MQTT=OK

```
16:08:01.306 ->
16:08:01.306 -> [U1 CH_CONFIG] Req dari 0x0001 RSSI:-109.0dBm
16:08:01.306 -> [CH_CONFIG] Respond ke 0x0001 (delay 400ms)
16:08:01.876 -> [CH_CONFIG] Resp ke-1
16:08:02.647 -> [CH_CONFIG] Resp ke-2
16:08:07.091 -> [U1 CH_HELLO] CH=0x0001 RSSI:-109.0dBm
16:08:07.091 -> → cluster=0x0001 parent=0x0002 dicatat
16:08:24.928 -> [HB|330s] rx=5 ok=0 WiFi=OK MQTT=OK
16:08:54.906 -> [HB|360s] rx=5 ok=0 WiFi=OK MQTT=OK
```

GW mencatat: CH1 terhubung via CH2
Topologi: CH1 → CH2 → CH3/GW

Topologi Normal – Data Flow Berhasil



```
16:10:22.146 ->
16:10:22.146 -> +-----+
16:10:22.146 -> | CH_DATA_UP #21 dari CH=0x0002 (radio U3)
16:10:22.180 -> | RSSI:-103.0dBm SNR:8.0dB seq:5
16:10:22.180 -> +-----+
16:10:22.180 -> | Node:0xAA01 Temp:24.0C Hum:42.0% Batt:3247mV Light:478
16:10:22.180 -> | Sukses: 5/21
16:10:22.180 -> +-----+
16:10:22.180 -> [MQTT] Publish OK
16:10:22.180 -> [JSON] {"gatewayId":3,"radioSrc":3,"chId":2,"nodeId":43521,"temp":24,"hum":42,"battMv":3247,"light":478,"rssi":-103,"snr":8,"seq":5,"rxOk":1
16:10:24.901 -> [HB/450s] rx=21 ok=5 WiFi=OK MQTT=OK
16:10:27.164 ->
16:10:27.164 -> +-----+
16:10:27.164 -> | CH_DATA_UP #24 dari CH=0x0002 (radio U3)
16:10:27.164 -> | RSSI:-103.0dBm SNR:8.2dB seq:6
16:10:27.164 -> +-----+
16:10:27.164 -> | Node:0xAA01 Temp:32.0C Hum:72.0% Batt:3143mV Light:166
16:10:27.164 -> | Sukses: 6/24
16:10:27.164 -> +-----+
16:10:27.208 -> [MQTT] Publish OK
16:10:27.208 -> [JSON] {"gatewayId":3,"radioSrc":3,"chId":2,"nodeId":43521,"temp":32,"hum":72,"battMv":3143,"light":166,"rssi":-103,"snr":8.25,"seq":6,"rxOk":1
```

Skenario 2

Failover

CH2 Mati → CH1 Mencari Parent Baru



STEP 7

CH2 OFF → ACK Timeout (3x) → Failover!

Data Masuk dari Node

0xAA01 RSSI:-113dBm, Temp, Hum, Batt...

TX → 0x0002 attempt 1/3

ACK Timeout, backoff 100ms

TX → 0x0002 attempt 2/3

ACK Timeout, backoff 200ms

TX → 0x0002 attempt 3/3

ACK Timeout, backoff 400ms

[FAIL] failCnt=3

→ [FAILOVER] CH_CONFIG broadcast!

CH_CONFIG ulang

Mencari parent terbaik @ 921MHz...

```
16:11:30.369 ->
16:11:30.369 -> [STAR RX] node=0xAA01 RSSI:-113.0dBm SNR:2.5dB
16:11:30.369 -> Temp:35.0C Hum:77.0% Batt:3374mV Light:1014
16:11:30.507 -> [MESH TX] node=0xAA01 seq=17 -> 0x0002
16:11:30.507 -> [TX-0x0002 attempt 1/3]
16:11:32.862 -> [ACK] Timeout attempt 1, backoff 100ms
16:11:32.955 -> [TX-0x0002 attempt 2/3]
16:11:35.330 -> [ACK] Timeout attempt 2, backoff 200ms
16:11:35.517 -> [TX-0x0002 attempt 3/3]
16:11:37.888 -> [ACK] Timeout attempt 3, backoff 400ms
16:11:38.279 -> [FAIL] parent=0x0002 failCnt=3
16:11:38.279 -> [FAILOVER] CH_CONFIG broadcast!
16:11:39.267 ->
16:11:39.267 -> [CH_CONFIG] Mencari parent terbaik @ 921MHz...
16:11:39.267 -> [CH_CONFIG] Broadcast 1/5...
16:11:39.715 -> [CH_CONFIG] Listening response 5 detik...
16:11:40.314 -> [CH_CONFIG] Response 0x0003 RSSI:-105.0dBm
16:11:41.080 -> [CH_CONFIG] Response 0x0003 RSSI:-105.0dBm
16:11:44.751 -> [CH_CONFIG] SELESAI: parent=0x0003(-105.0dBm) alt=0x0000
16:11:44.751 -> [CH_CONFIG] Parent berubah: 0x0002 -> 0x0003
16:11:44.751 -> [PREFS] Saved parent=0x0003 alt=0x0000
16:11:45.017 -> [CH_HELLO] -> 0x0003
16:11:45.513 -> [CH_HELLO] Terkirim
16:11:45.513 -> [FAILOVER] Retry kirim dengan parent baru...
16:11:45.513 -> [TX-0x0003 attempt 1/3]
16:11:47.882 -> [ACK] Timeout attempt 1, backoff 100ms
16:11:48.012 -> [TX-0x0003 attempt 2/3]
16:11:48.617 -> [ACK] OK seq=17
```

STEP 8

CH3/GW Merespon Re-Config CH1

Event: [U1 CH_CONFIG] Req dari 0x0001

RSSI: -108.0dBm

Aksi: Respond ke 0x0001 (delay 400ms)

Resp ke-1: @ 16:11:40.329

Resp ke-2: @ 16:11:41.080

CH_HELLO: CH=0x0001 RSSI:-108.0dBm

Dicatat: cluster=0x0001 parent=0x0003 ✓

16:11:39.716 -> [U1 CH_CONFIG] Req dari 0x0001 RSSI:-108.0dBm

16:11:39.760 -> [CH_CONFIG] Respond ke 0x0001 (delay 400ms)

16:11:40.329 -> [CH_CONFIG] Resp ke-1

16:11:41.080 -> [CH_CONFIG] Resp ke-2

16:11:45.513 -> [U1 CH_HELLO] CH=0x0001 RSSI:-108.0dBm

16:11:45.513 -> → cluster=0x0001 parent=0x0003 dicatat

16:11:48.617 ->

16:11:48.617 -> |-----|

CH3/GW kini menjadi parent CH1

menggantikan CH2 yang sedang offline.

Parent berubah: 0x0002 → 0x0003

Parent CH1 Berubah: CH2 → CH3/GW

Parent Lama: CH2 (0x0002)

OFFLINE
ACK Timeout 3x → failCnt=3

↓ **FAILOVER**

✓ Parent Baru: CH3/GW (0x0003)

ONLINE
RSSI: -105.0dBm | ACK OK! seq=17

[CH1 Log - Failover]

```
[FAIL] parent=0x0002 failCnt=3  
[FAILOVER] CH_CONFIG broadcast!  
Resp 0x0003 RSSI=-105.0dBm (x2)  
[SELESAI] parent=0x0003(-105dBm)  
Parent berubah: 0x0002 → 0x0003  
[CH_HELLO] → 0x0003 → Terkirim  
[ACK] OK seq=17 ✓
```

Perubahan Topologi:

SEBELUM: Node → CH1 → CH2 → GW

SESUDAH: Node → CH1 —————→ GW

(melewati CH2, langsung ke CH3/GW)

Data terkirim, tidak ada loss! ✓

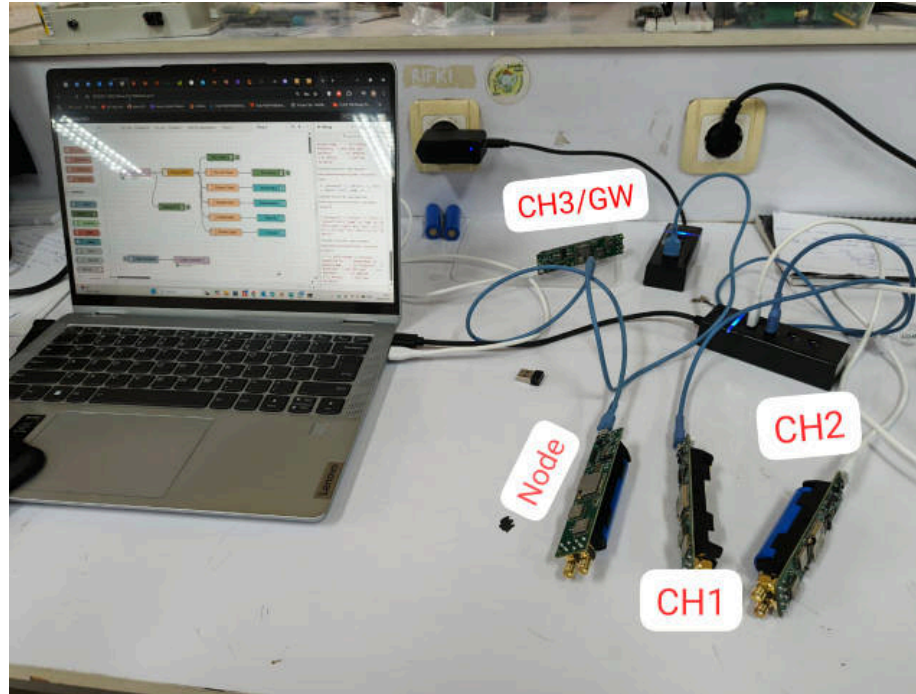
Topologi Baru Setelah Failover – Data Flow



← Jalur Baru (Failover Path) →

```
6:15:03.382 -> [U3_CH_CONFIG] Req dari 0x0002 RSSI:-110.0dBm
6:15:03.382 -> [CH_CONFIG] Respond ke 0x0002 (delay 400ms)
6:15:03.968 -> [CH_CONFIG] Resp ke-1
6:15:04.733 -> [CH_CONFIG] Resp ke-2
6:15:08.045 ->
6:15:08.045 -> +-----+
6:15:08.045 -> | CH_DATA_UP #102 dari CH=0x0001 (radio UI)
6:15:08.045 -> | RSSI:-111.0dBm SNR:3.8dB seq:58
6:15:08.045 -> +-----+
6:15:08.045 -> | Node:0xAA01 Temp:28.0C Hum:64.0% Batt:3266mV Light:591
6:15:08.092 -> | Sukses: 56/102
6:15:08.092 -> +-----+
6:15:08.092 -> [MQTT] Publish OK
6:15:08.092 -> [JSON] {"gatewayId":3,"radioSrc":1,"chId":1,"nodeId":43521,"temp":28,"hum":64,"battMv":3266,"light":591,"rsi":-111,"snr":3.75,"seq":58,"rxC
6:15:09.180 -> [U3_CH_HELLO] CH=0x0002 RSSI:-111.0dBm
6:15:13.062 ->
6:15:13.062 -> +-----+
6:15:13.062 -> | CH_DATA_UP #104 dari CH=0x0001 (radio UI)
6:15:13.062 -> | RSSI:-110.0dBm SNR:3.8dB seq:59
6:15:13.062 -> +-----+
6:15:13.062 -> | Node:0xAA01 Temp:31.0C Hum:78.0% Batt:3415mV Light:349
6:15:13.062 -> | Sukses: 57/104
```

Setup Fisik Pengujian



Kesimpulan

Pengujian Cluster Head – Gas Leak Detector

Konfigurasi Otomatis

Semua CH (CH1, CH2, CH3/GW) berhasil melakukan discovery dan memilih parent berdasarkan RSSI terbaik secara otomatis.

Topologi Star-Mesh Berfungsi

Data berhasil dikirim Node → CH1 → CH2 → CH3/GW dengan ACK confirmation setiap hop.

Failover Otomatis Sukses

Saat CH2 offline, CH1 otomatis deteksi timeout 3x, broadcast ulang, dan pindah parent ke CH3/GW.

Re-routing Transparan

Setelah failover, data tetap terkirim via jalur baru Node → CH1 → CH3/GW tanpa intervensi manual.

Uji sistem di lingkungan kampus



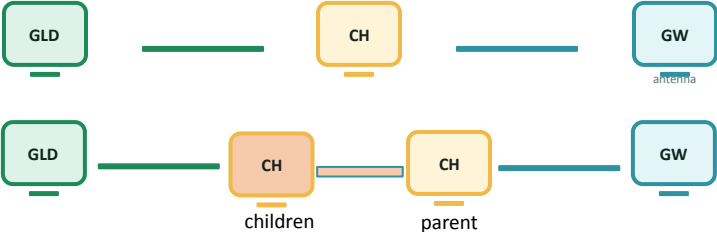
No	Kondisi / Lingkungan	Jarak (m)	Tinggi Ant. Tx (m)	Tinggi Ant. Rx (m)	RSSI (dBm)	PDR
1	area terbuka	10	3	3	-47	100%
2	area terbuka	30	3	3	-57	100%
3	area terbuka	100	3	3	-71	100%
4	Bawah pepohonan	80	3	3	-78	100%
5	bawah pepohonan	80	4	4	-84	100%
6	belakang gedung + pepohonan	-	3	3	-93	98%
7	belakang gedung + pepohonan (ant. diubah)	-	2	6	-81	100%
8	area Indomaret	-	2	6	-104	68%
9	area Fisika (batas jangkauan)	-	2	6	-112	26%

Pengaturan ketinggian antena berpengaruh terhadap kemampuan komunikasi data, namun perlu ditinjau kembali kondisi lingkungan, jika banyak vegetasi, mtingkatkan tinggi antena TX hingga mendekati dedaunan dan cabang pohon menyebabkan RSSI turun. Namun jika antena TX diturunkan, dan antena RX dinaikkan hingga 6 m, kemampuan komunikasi menjadi lebih baik (area Gedung dan pepohonan)

Pengujian simulasi komunikasi LoRa di sekitar ITB

Simulasi dilakukan sebelum site survey agar tim punya baseline jarak, kualitas link, failover, dan ketahanan CH outdoor.

Topologi uji: GLD -> CH -> GW



PDR

packet received / sent

RSSI

kekuatan sinyal

SNR

kualitas sinyal vs noise

Catatan: titik uji ITB dibuat mewakili skenario indoor, indoor dengan obstacle pohon/gedung, dan outdoor terbuka.

1. Uji paket data dan jarak

GLD dummy beberapa titik. Catat jarak GLD-CH-GW, RSSI, SNR, PDR, latency, dan obstacle.

2. Uji failover CH

Matikan beberapa CH. Pastikan A-CH mencari parent baru dan data tetap sampai ke GW.

3. Uji daya CH outdoor

Nyalakan CH dengan solar/baterai, enclosure outdoor, dan logging minimal satu minggu.

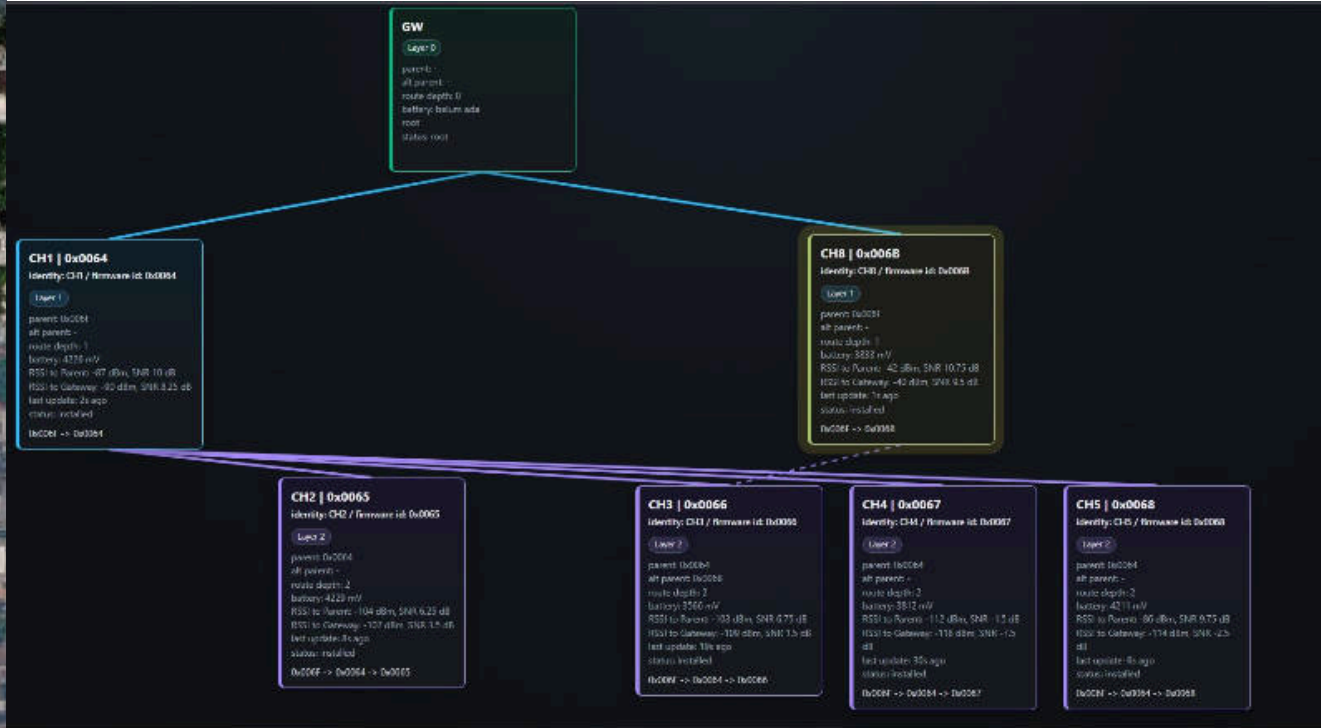
Parameter yang dicatat per test point

- Koordinat TX/RX dan elevasi.
- LOS/NLOS, obstacle gedung/pohon/beton.
- Konfigurasi radio: SF, BW, TX power, antenna.
- Packet sent, received, PDR, RSSI, SNR, retry.

Uji Mesh Cluster Head

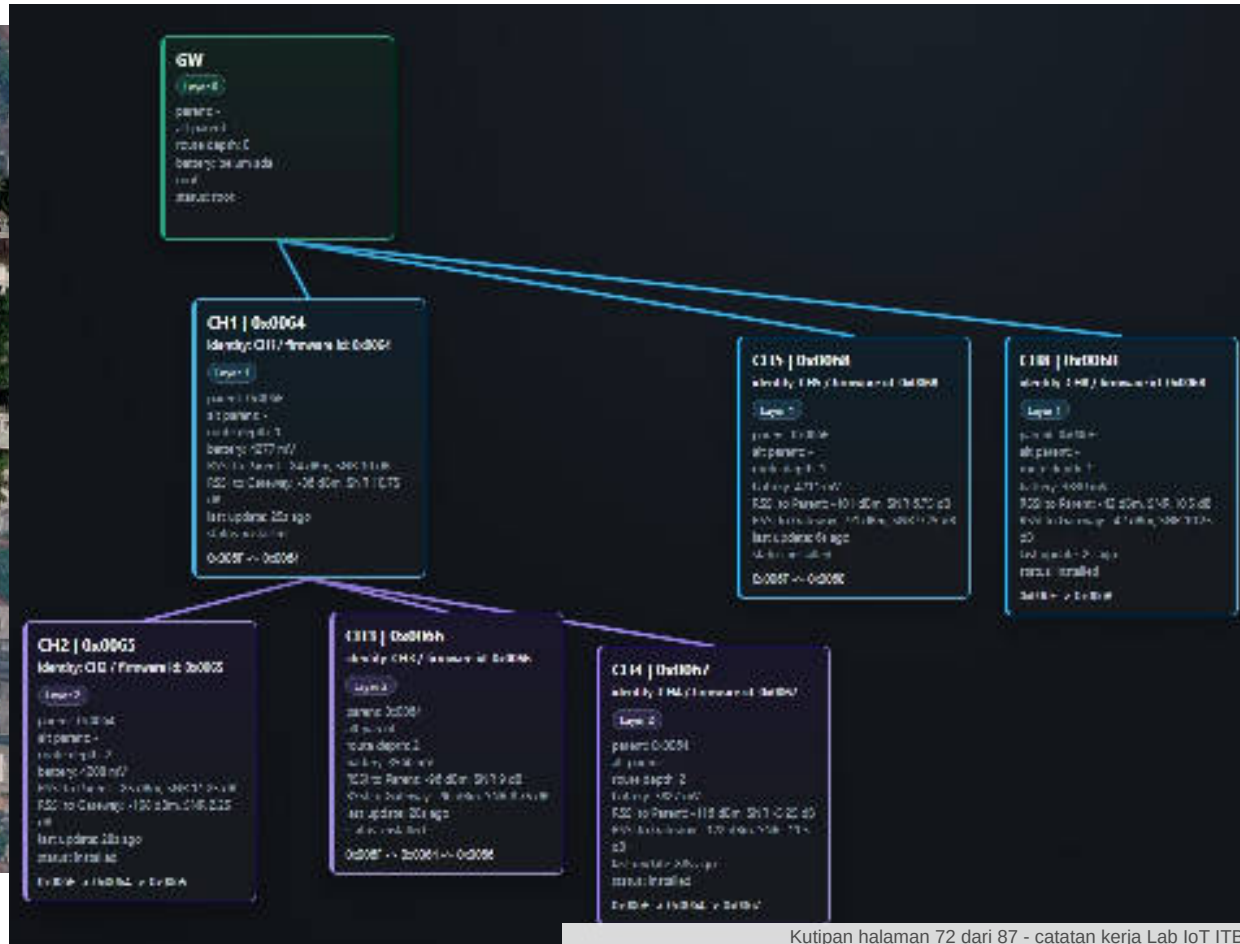
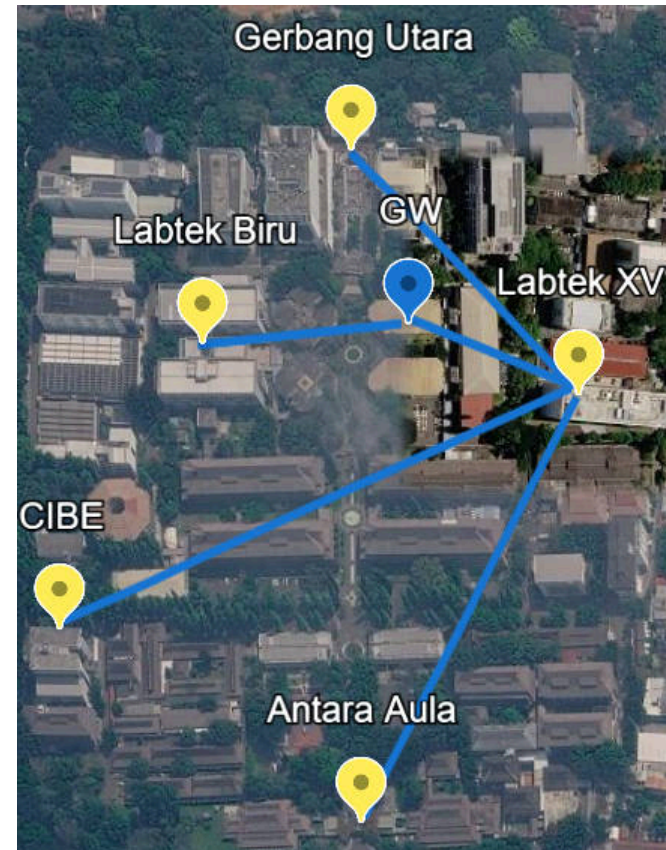
Uji Mesh 1

16 Juli 2026



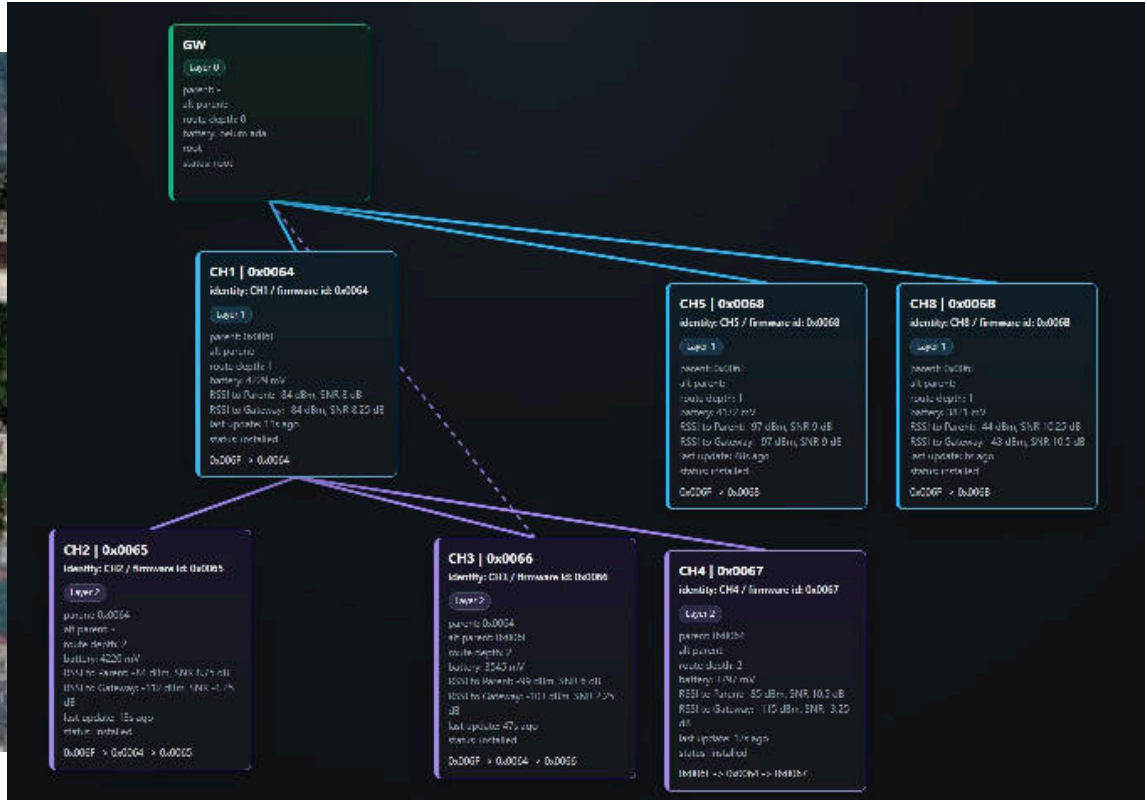
Uji Mesh 2

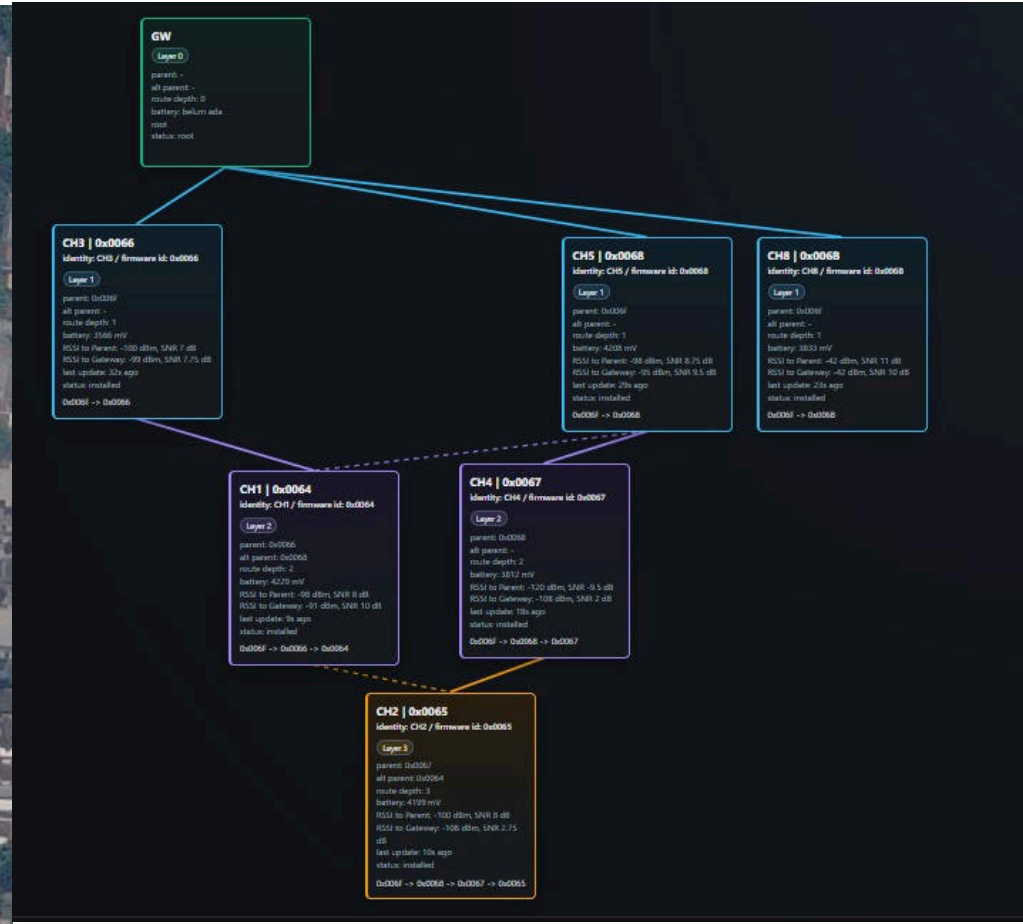
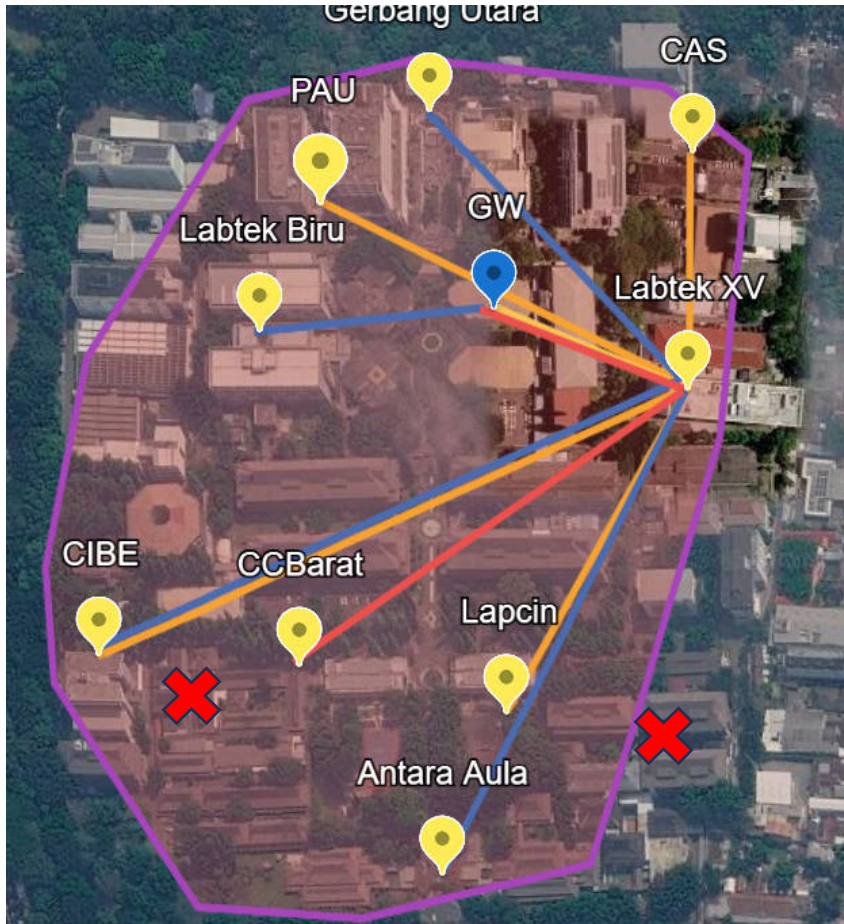
16 Juli 2026



Uji Mesh 2

16 Juli 2026





Uji Downlink GLD 1

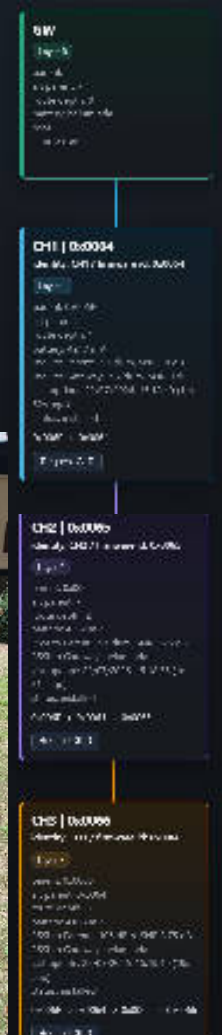


Uji Downlink GLD 2

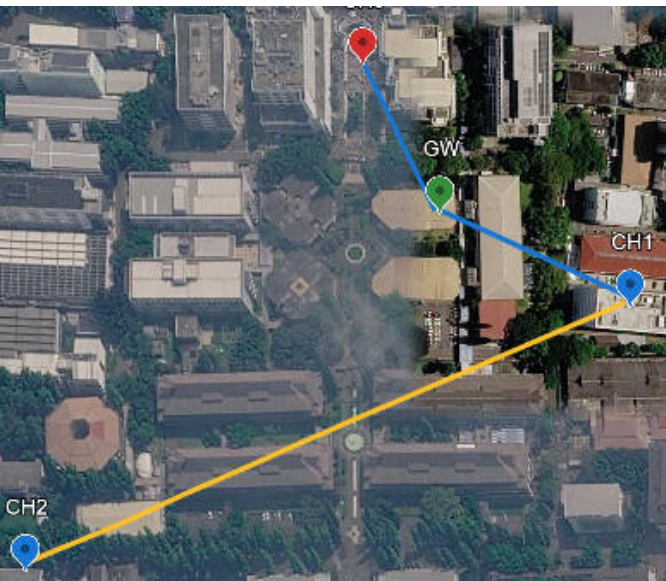


GW Gateway ID: GW0001 Location: 10.111111, 10.111111 Status: Online	
CH1 0x0004 Channel 1 (Node 4) ID: CH10004 Location: 10.111111, 10.111111 Status: Online	
CH2 0x0005 Channel 2 (Node 5) ID: CH20005 Location: 10.111111, 10.111111 Status: Online	
CH3 0x0006 Channel 3 (Node 6) ID: CH30006 Location: 10.111111, 10.111111 Status: Online	

Uji Downlink GLD 3

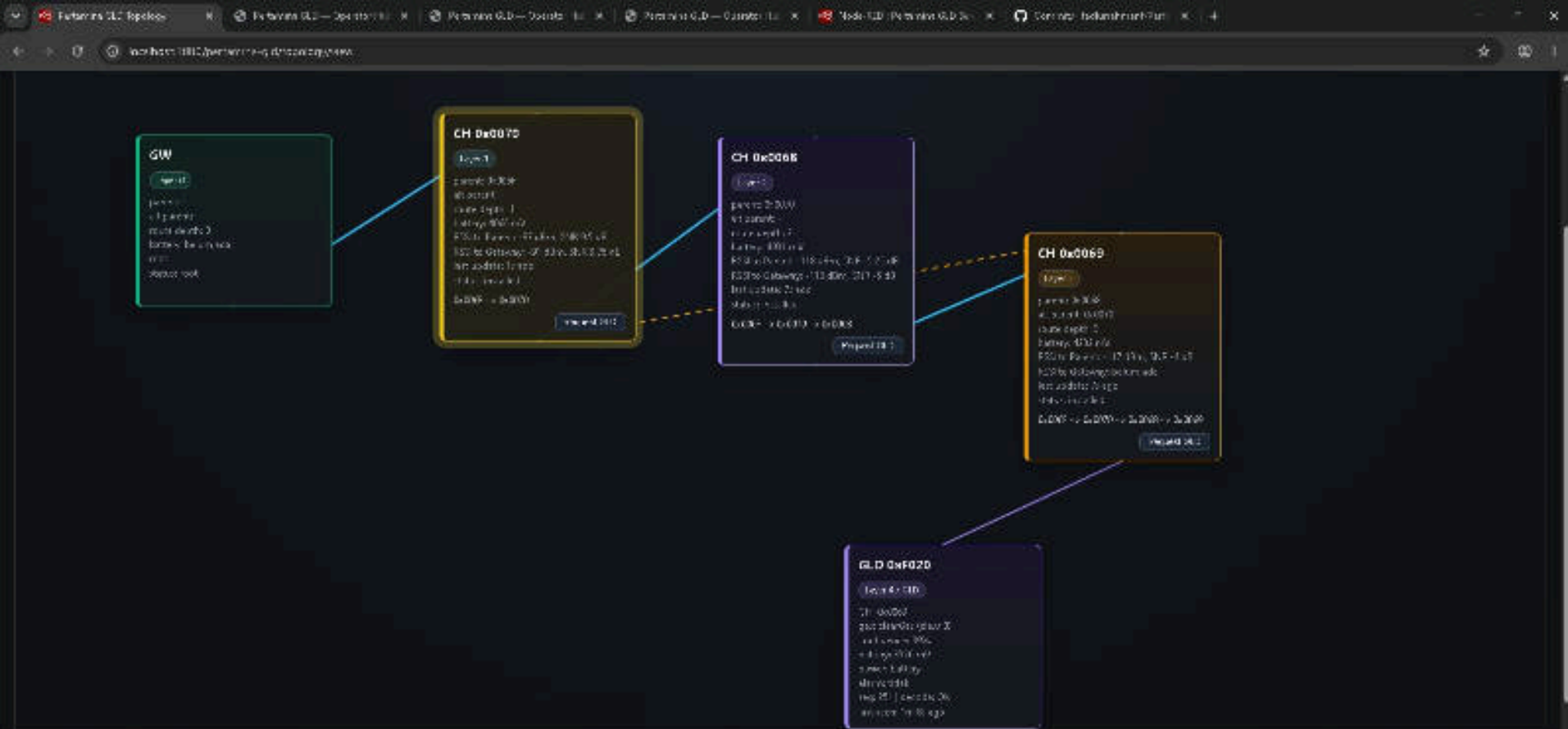


Uji Downlink GLD 4



Analisis dan Perbaikan Firmware Downlink GLD





Status CH

Open: 1/1/1

CH	Status	Parent	Role	Last CH GLD	Expected Next Event	Act	Update
CH 0x0070	init	0x0000	0x0000	11 ago	Expected Next Event: 0x0000	Refresh	11 ago
CH 0x0065	init	0x0000	0x0000	11 ago	Expected Next Event: 0x0000	Refresh	11 ago